1.3 Conditional Probability

Conditional probability

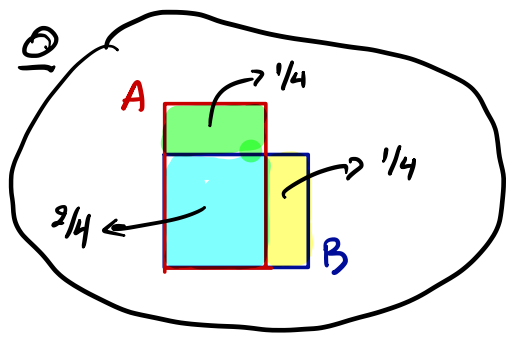
*The **conditional probability** of event A given that event B has occurred is*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Example

available known information

Find $\mathbb{P}(A|B)$.



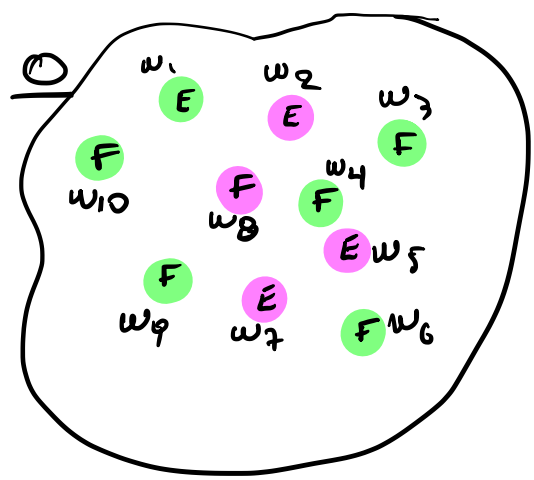
$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

$$= \frac{2/4}{3/4} = \frac{2}{3}$$

Example

Consider the previous example, and let's assume that some of the spheres are empty (E) and some are filled (F)

If we pick a green sphere, what is the probability of being empty?



$\mathbb{P}(E/G) = ?$

$E = \{\omega_1, \omega_2, \omega_5, \omega_7\}$

$F = \{\omega_3, \omega_4, \omega_6, \omega_8, \omega_9, \omega_{10}\}$

$$\left.\begin{array}{l} \mathbb{P}(E|G) = \dfrac{\mathbb{P}(E\cap G)}{\mathbb{P}(G)} \\ E\cap G = \{\omega_1\} \Rightarrow \mathbb{P}(E\cap G) = \frac{1}{10} \end{array}\right\} \Rightarrow \mathbb{P}(E|G) = \frac{1/10}{6/10} = \frac{1}{6}.$$

Multiplicative rule

For a sequence of events A, B, and C, we have

$$\mathbb{P}(A\cap B\cap C) = \mathbb{P}\left((A\cap B)\cap C\right) = \mathbb{P}(A\cap B)\mathbb{P}(C|A\cap B)$$

or

$$\mathbb{P}(A\cap B\cap C) = \mathbb{P}(A)\mathbb{P}(A|B)\mathbb{P}(C|A\cap B).$$

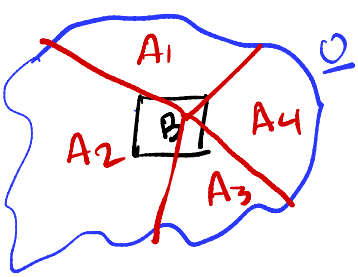
This rule can be generalized for any sequence.

Partition

*A **partition** of a sample space Ω is a collection of events A_n such that*

$$\bigcup_n A_n = \Omega$$

where $A_n \cap A_m = \emptyset, n \neq m$.



Clearly:

$$B = \cup_n [B\cap A_n]$$

Using the third axiom, we get:

$$\mathbb{P}(B) = \sum_n \mathbb{P}(B\cap A_n)$$

Independence

Two events A and B are ***independent*** *if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Consequently, from the definition of independence, we have

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A).$$